# Ali Atyabi

Mechanical Engineer, Ph.D. | CFD modeling of mass transport phenomena in Membrane-based processes

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## Profile Summary

Ph.D. mechanical engineer with 10+ years of specialized expertise in CFD modeling of transport phenomena in membrane-based separation processes. Proven track record in developing advanced computational models for electrodialysis, membrane chromatography, osmotic pump, and electrochemical systems. Experienced in mentoring students, collaborating with industry partners, and delivering practical solutions to complex modeling challenges. Strong communicator with hands-on approach to research and excellent project management skills. Currently at KU Leuven with deep understanding of academic research environment and industrial collaboration.

## Core Technical Expertise

**CFD modeling:** ANSYS Fluent, COMSOL Multiphysics
**Membrane separation processes:** Electrodialysis, forward osmosis, membrane chromatography, ion exchange membranes, membrane humidifiers
**Programming, visualization, and automation:** Python (NumPy, Pandas, Matplotlib, SciPy), post-processing automation, data analysis, parametric studies, process optimization
**Machine Learning and optimization:** Genetic algorithms, design of experiments (DOE), familiar with ML concepts for model reduction
**Engineering design:** SolidWorks, AutoCAD
**Research and collaboration:** Project management, technical writing, student mentoring

## Languages

English (Fluent), Dutch (A1.2, improving), Persian (Native)

## Education

**Ph.D. in Mechanical Engineering** 2014–2019
*Energy Conversion*, University of Isfahan, Iran
Focus: Electrochemical membrane-based systems, mass transport phenomena, thermodynamics, optimization

**M.Sc. Mechanical Engineering** 2010–2013
*Energy Conversion*, Isfahan University of Technology, Iran
Specialization: Flow field design, mass transport, multiphysics model

**B.Sc. Mechanical Engineering** 2005–2009
*Solid Design*, Isfahan University of Technology, Iran

## Training

**EIS Modeling Training** Jul 2023
Luzerne, Switzerland

## Key Achievements & Impact

- Successfully supported 10+ industry-academic research projects combining CFD expertise with practical problem-solving
- Trained and mentored 4+ graduate students in CFD techniques, transport modeling, and simulation methodologies
- Established productive collaborations with major industry partners including imec and John Cockerill
- Delivered technical presentations explaining complex transport phenomena to diverse audiences
- Currently working at KU Leuven with established understanding of department culture and research objectives

## Professional Experience

**KU Leuven university** 2024–Present
*Researcher, Belgium*
- Developed comprehensive CFD models for electrodialysis membrane systems using COMSOL Multiphysics, incorporating coupled species transport, and electrochemical reactions
- Conducted parametric CFD studies to optimize membrane-based separation processes, analyzing flow distribution, concentration polarization effects, and membrane performance metrics
- Developed comprehensive CFD models for an osmotic pump using RO membrane for medical applications
- Mentored and trained PhD students on CFD modeling techniques, transport phenomena theory, and simulation best practices for membrane systems
- Collaborated with multidisciplinary research teams on projects combining experimental and computational approaches

**VITO** 2021–2024
*R&D Researcher, Belgium*
- Designed and optimized flow field configurations for alkaline water electrolyzers using multiphase CFD simulations in COMSOL, achieving 30% energy reduction through improved mass transport of bubbles
- Developed detailed CFD models for porous transport layers and ion exchange membrane systems, reducing operational voltage enhanced species and gas transport modeling
- Created Python-based tools for dynamic process modeling, automated mass balance calculations, performance prediction, and parametric analysis of electrochemical membrane systems
- Managed collaborative research projects with John Cockerill, providing technical support for membrane selection, characterization, and industrial-scale system design
- Performed CFD simulations of manifolds, flow distributors, and membrane assemblies for scale-up to industrial applications

**INSF - Iran National Science Foundation** 2020–2021
*Researcher, Tehran*
- Developed advanced CFD models for membrane electrode assemblies with coupled heat and mass transfer in PEM fuel cell system
- Collaborated with Braunschweig University (Germany) on international research projects focused on CFD modeling of mass transport phenomena in PEM fuel cell system

**Persian Gulf university, Boushehr** 2019–2020
*Researcher*
- Developed CFD models for coke injection in ethylen-ethan cracking furnace.
- Optimized oxygen mixture nozzle sparger designs using CFD.
- Evaluated centrifugal/Lobe and rotary high-pressure piston pumps designs post-modification

**University of Isfahan** 2014–2019
*PhD Researcher, Isfahan*
- Conducted comprehensive CFD modeling of high-temperature membrane systems, analyzing multi-component transport phenomena across ion-conducting membranes
- Designed and optimized membrane humidifier systems using integrated computational and experimental approaches
- Published peer-reviewed research on transport modeling in membrane processes and separation systems
- Demonstrated initiative in developing novel modeling approaches for complex membrane applications